THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 5 & 6: **Describing Motion Around Us, How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A runner completes one full lap of a 400 m track. Her displacement is: 1

(a) 400 m **(b)** 800 m **(c)** zero **(d)** 200 m

2. The slope of a velocity–time graph gives: 1

(a) acceleration **(b)** velocity **(c)** speed **(d)** displacement

3. Which equation is NOT a kinematic equation for constant acceleration? 1

(a) $v^2 = u^2 + 2as$ **(b)** $v = u + at$ **(c)** $s = ut + \frac{1}{2}at^2$ **(d)** $v = s/t^2$

4. The acceleration of a freely falling body near Earth's surface is about: 1

(a) 98 m s^{-2} **(b)** 0.98 m s^{-2} **(c)** 1.6 m s^{-2} **(d)** 9.8 m s^{-2}

5. A body in uniform circular motion of radius r completes one revolution in time T. Its average speed is: 1

(a) r/T **(b)** zero **(c)** $2\pi r/T$ **(d)** $\pi r/T$

6. Friction acting on a moving object is directed: 1

(a) perpendicular to the surface only **(b)** along its motion
(c) opposite to its motion **(d)** downwards

7. A gun recoils when a bullet is fired. This is explained by: 1

(a) Hooke's law **(b)** Newton's third law
(c) law of gravitation **(d)** Newton's first law

8. We can walk because the ground exerts a _____ on our feet in the forward direction. 1

(a) buoyant force **(b)** magnetic force
(c) force of friction **(d)** tension

9. Galileo's thought experiment on a smooth horizontal plane suggested that a body: 1

(a) moves in a circle
(b) always needs a force to move
(c) keeps moving indefinitely if no friction acts
(d) stops immediately

10. The momentum of a 2 kg ball moving at 5 m s^{-1} is: 1

(a) 25 kg m s^{-1} **(b)** 7 kg m s^{-1} **(c)** 2.5 kg m s^{-1} **(d)** 10 kg m s^{-1}

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Give one example each of uniform and non-uniform motion in a straight line. 2

12. A ball falls from rest for 2 s. Find its velocity ($g = 10\ m\ s^{-2}$). 2

13. How does a rocket move upward? 2

14. A bullet of mass 20 g is fired from a gun. Why is the recoil of the gun much smaller in velocity than the bullet's velocity? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Using the Ganitakaumudi example, two postmen start 210 yojanas apart and walk towards each other at 9 and 5 yojanas per day. When do they meet? 3

16. A body of mass 4 kg changes its velocity from 6 m s^{-1} to 14 m s^{-1} in 4 s. Find the force. 3

17. A body travels 20 m in the first 2 s and 40 m in the next 2 s in a straight line. Is the motion uniform? Find average speeds. 3

OR

A motorbike with u = 28 m s^{-1} stops after 98 m. Find the acceleration and time. 3

18. Why do vehicles' tyres have treads and why is driving on a wet road risky? 3

OR

Explain what happens if the force of friction disappears from the world. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Two cars A and B start together; on a position–time graph A is a line of slope 10 m s^{-1} and B of slope 20 m s^{-1}. 5

(i) Which is faster?
(ii) Find the position of each after 6 s.
(iii) After how many seconds is B 100 m ahead of A?
(iv) What would a horizontal line mean?

OR

(i) State the conditions under which the kinematic equations are valid. 5
(ii) A motorbike with u = 28 m s^{-1} stops after 98 m. Find the acceleration.
(iii) Find the time taken.
(iv) Why is the sign of a negative?

20. (i) State Newton's second law in terms of momentum. 5
(ii) Define momentum.
(iii) A 2 kg ball moving at 5 m s^{-1} is stopped in 0.5 s; find the force.

OR

(i) State the three laws with an example each. 5
(ii) Which law explains why seat belts are essential?

END OF QUESTION PAPER — **Total: 40 Marks**